

Single View Might Be Enough: A Task-Dependent Study of Multi-Perspective Necessity in Geosensing

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Abstract. Pairing overhead (satellite) and street-level imagery is widely assumed to help vision-language models reason about cities. We test this assumption directly, building a multi-city visual question answering benchmark of 40 cities in which each location carries a satellite tile and four street-level views, with four-way multiple-choice questions about eight urban attributes derived from OpenStreetMap. A controlled image ablation—no image, satellite only, street only, and both—lets us measure how much each view contributes. For these attribute tasks the two-view condition exceeds the better single view by only about two to three points on average, while on a built-in set of cross-view control questions, which require both views by construction, the same model gains 17 to 70 points. A paired item-level analysis shows the second view is not merely redundant but mildly harmful: on the items it changes, it overturns a correct answer about ten times as often as it rescues a wrong one, so that even an oracle keeping the better single view per question beats the two-view model. The pattern holds across a 9B model we fine-tune on the data, four off-the-shelf remote-sensing VLMs, two city splits, and three prompting strategies including extended chain-of-thought. Multi-perspective necessity is thus task-dependent: for urban-attribute recognition at commonly served resolutions, one well-chosen view is competitive with two.

Keywords: Geosensing · Vision-language models · Multi-view fusion

1 Introduction

A recurring premise in geospatial machine learning is that more perspectives are better. Overhead imagery captures layout, footprints, and connectivity; ground imagery captures facades, surfaces, signage, and street furniture. Pairing the two is intuitively complementary, and a growing line of work collects, aligns, and jointly models satellite and street-level views for urban understanding.

We ask whether this expectation holds for concrete urban-attribute recognition, and find that for the tasks we study it largely does not. We construct a multi-city visual question answering (VQA) benchmark in which every location is represented by a marked satellite tile and four cardinal street-level views, and pose four-way multiple-choice questions about eight urban attributes (land use,

One held-out location (Sydney): each single view already answers the question; the second view adds little.

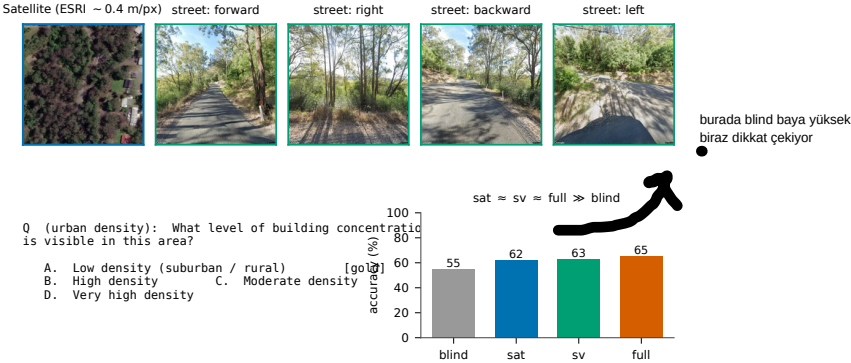


Fig. 1: The benchmark and the finding on one held-out location (Sydney). Each location pairs a marked sub-meter satellite tile with four cardinal street-level views; questions ask about urban attributes (here, urban density). The ablation (bottom right, trained Qwen3.5-9B on this task) shows the two-view condition (full) barely exceeds the better single view (sat $\approx$ sv), while both far exceed the no-image prior (blind): for this task the second view is largely redundant. Imagery is unmodified dataset content: satellite tile © Esri World Imagery; street-view panels © Google (Street View), shown with their original attribution.

building height, urban density, road type, road surface, junction type, amenity richness, and transit density) grounded in OpenStreetMap (OSM) metadata [19]. Crucially, the same imagery also supports a family of *cross-view* questions that are unanswerable from a single view by construction (e.g. “does this street-level photo belong to this satellite tile?”). These cross-view tasks serve as a built-in positive control. Figure 1 illustrates the setup and the headline effect on a single location.

Our central instrument is a controlled image ablation. For each question we evaluate a model under four input conditions—blind (no image), sat (satellite only), sv (street only), and full (both)—and define *fusion synergy* as $\text{full} - \max(\text{sat}, \text{sv})$: the accuracy gained from having both views over the better single view. Positive synergy is the signature of genuine cross-view fusion. We complement synergy with *vision contribution*, $\text{full} - \text{blind}$, which isolates how much the imagery contributes at all over a text-only prior.

On the eight attribute tasks, having both views improves on the better single view by only two to three points on average—for a 9B model we fine-tune on the benchmark, for four off-the-shelf remote-sensing VLMs evaluated zero-shot, for two city splits, and for three prompting strategies including a 16k-token reasoning budget. A paired item-level analysis (Sec. 5.5) shows the second view is mildly harmful rather than merely redundant: on the items where it changes the answer it overturns a correct one about ten times as often as it rescues a wrong one (10% versus 1%; McNemar significant). As a result, even an oracle that keeps

the better single view per question (an upper bound, not an achievable system) beats the two-view model by 8–9 points. Vision still matters (full – blind $\approx +14$ for the fine-tuned model): the model reads the imagery, but one view supplies about as much usable signal as two. On the cross-view controls the same model gains 17–70 points, so the ablation detects fusion when the task requires it.

This is a *task-dependent* observation about single-view-decidable attribute recognition at commonly served satellite resolutions, not a claim that overhead imagery is uninformative or that multi-view fusion is futile in general. We report bootstrap confidence intervals on every synergy estimate, audit the benchmark for text-only leakage, and characterize the items where the second view changes the answer.

Our contributions are: (i) a multi-city satellite + street VQA benchmark (40 cities, eight OSM-grounded attribute tasks plus cross-view controls) with a location-level held-out split and a four-way image ablation; (ii) the measurement that the second view’s contribution to these tasks is small and robust across model scale, training regime, city split, and prompting strategy, with the cross-view controls confirming the measurement is sensitive; (iii) a paired item-level analysis showing the second view overturns about ten correct answers for each one it rescues, so an oracle single-view router beats the two-view model; and (iv) a reusable ablation protocol (synergy, vision-contribution, interference asymmetry) for deciding when a second view is warranted.

2 Related Work

Remote-sensing vision-language models are single-view. A wave of instruction-tuned VLMs target remote sensing, including GeoChat [7], SkySenseGPT [16], LHRS-Bot and LHRS-Bot-Nova [10, 18], EarthGPT [29], and the multi-sensor generalist EarthDial [21]. These models are overhead-only: each consumes a single aerial or satellite image, and none ingests ground-level imagery (Tab. 2). Where “multi-modal” appears in this line of work it refers to multiple *overhead* sensors—optical, SAR, and infrared, as in EarthGPT—rather than to pairing satellite with street-level views. The dominant evaluation suites follow suit: RSVQA [15], VRSBench [9], and GeoChat’s own benchmark all pose questions over a single overhead image. None isolates the marginal value of a second, ground-level view, which is the question we study. Architecturally these models build on LLaVA-style designs [12, 13] with CLIP or SigLIP encoders [20, 28]; we use four as zero-shot baselines and discuss their training-data provenance in Sec. 4.

Multi-view urban benchmarks. A smaller set of benchmarks pairs overhead and ground views for urban understanding, and two report observations that converge with ours. UrBench [31] evaluates multimodal LLMs on coordinate-aligned street and satellite pairs and finds little difference between multi-image and single-image performance, attributing the gap to question comprehension rather

than fusion. The concurrent CityLens [14] benchmarks LVLs on socioeconomic-indicator regression from satellite and street-view and reports that street view alone performs comparably to using both views and far above satellite alone. That two independent studies see the same redundancy strengthens our premise. These works do not, however, provide the per-item result at the center of our paper: CityLens runs a three-task, single-model modality comparison on a *regression* target, and neither isolates how the second view flips individual items or whether an oracle single-view router beats the two-view model. We also add cross-view control tasks—absent from prior urban benchmarks—that show the same instrument registers large fusion when fusion is required. UrbanLLaVA [3] trains over satellite, street-view, OSM, and trajectory inputs but poses its attribute-style tasks from satellite alone and reserves multi-view supervision for matching and localization; CityCube [26] targets cross-view spatial reasoning rather than attribute recognition.

Cross-view geo-localization: where both views are necessary. ayr bir section olarak datasette yazilabilir, yeri benchmark ksmna alınabilir Satellite ground pairing is well established in cross-view geo-localization, where the goal is to match a ground photo to an overhead reference [2, 27, 32]. There the two views are jointly *necessary*—one cannot localize the ground image without the overhead one—and methods succeed by modeling the distinct-but-linkable layout the views share [30]. This is the regime our cross-view controls occupy; complementarity that is constitutive of localization is largely absent from single-view-decidable attribute recognition.

The marginal value of a second modality. That adding a modality need not help is known beyond geosensing. Wang *et al.* [23] show the best single-modal network can outperform a naively fused one, and Liang *et al.* [11] formalize multimodal interaction into redundancy, uniqueness, and synergy, showing low-synergy regimes are real; our “fusion synergy” is the behavioral analogue of their synergy term. Against this, multimodal learning can provably lower achievable risk [5], and several aerial+ground systems report recognition gains [17, 25]. Most directly counter to our setting, Suel *et al.* [22] find that fusing satellite and street imagery improves *regression* of income and deprivation; we engage this in Sec. 7, noting it differs in task type (regression vs. four-way recognition), labels, and model (a trained CNN vs. our VLMs). Our finding refines rather than contradicts the “fusion helps” literature by naming a regime where it does not.

3 The Benchmark

Construction. The benchmark pairs overhead and street-level imagery with multiple-choice questions about urban attributes, generated from OpenStreetMap (OSM) metadata [19]. Each *location* is sampled within one of 40 cities across 32 countries on six continents and is associated with (i) one satellite tile, (ii) four street-level views at fixed bearings (forward, backward, left, right), and

(iii) OSM-derived attribute labels. Satellite imagery is sourced per-location by geography—sub-meter aerial orthophotos: ESRI World Imagery (~ 0.3 m/px to 0.5 m/px in cities) as the global default, NAIP (~ 0.6 m/px) in the US, and IGN (~ 0.2 m/px) in France, with a coarse Sentinel-2 fallback (~ 10 m/px) reserved for locations that no high-resolution source covers. In the benchmark every overhead tile resolves to a sub-meter source (4775 ESRI, 305 NAIP, 160 IGN; the Sentinel-2 fallback was never triggered), so the redundancy we report below is measured against genuinely high-resolution overhead imagery rather than a coarse proxy. The satellite tile carries a marker indicating the queried location; the marker is part of the input condition and is held fixed across all ablation modes (we examine its effect in Sec. 6). The four street-level views are retrieved from Google Street View and retain Google’s original attribution; satellite imagery is © its respective provider (Esri, NAIP/USDA, IGN).

Tasks. We study eight *attribute* tasks—`land_use`, `building_height`, `urban_density`, `road_type`, `road_surface`, `junction_type`, `amenity_richness`, and `transit_density`—each posed as a four-way multiple-choice question with a single correct letter. These are the questions for which “do we need both views?” is well-posed: each is, in principle, answerable from imagery of a single location. The dataset additionally contains *cross-view* tasks (view-view matching at two difficulty levels, in binary and multiple-choice form, and a camera-direction task) whose answer lives in the *relationship* between the two views; these are unanswerable from a single view by construction and we use them only as a positive control (Sec. 5.2).

Splits. All splits operate at the location level so that no location’s questions span two splits. We hold out 10% of locations *per city* as a benchmark test set; this set is identical across training regimes and is disjoint from training data (verified: zero question-id overlap). For training we use two strategies that share this benchmark: a *per-city* split (a further fraction of locations per city to validation) and a *seen/unseen* split (eight cities held out entirely to validation, to probe cross-city generalization). Question generation uses a fixed seed for option shuffling, and we keep a single question per (location, task) pair to avoid location-level inflation. Locations with fewer than four street-level views are discarded.

Leakage audit. Because the questions are templated, we audit how much of each task is solvable from text alone by evaluating every model in blind mode (no image) and treat full — blind as the vision-attributable signal. This audit flagged one further attribute task (green-space presence) that a model answered at 100% accuracy with no image, because the correct option wording was identifiable independent of its A/B/C/D position; we *exclude it entirely* from all results, leaving the eight tasks reported here. The retained tasks show no exploitable option-text or position regularity (Sec. 6).

Table 1: Benchmark test set (eight attribute tasks, held-out 10% of locations per city). Each question has four options and a single correct letter; image mode is a single marked satellite tile plus four street-level views.

Property	Value
Cities	40
Countries	32
Locations (test)	421
Questions (test)	2932
Attribute tasks	8
Options per question	4
Satellite views / location	1
Street-level views / location	4
<i>Questions per task</i>	
land_use / urban_density / road_type	421 each
amenity_richness / transit_density	421 each
junction_type	334
road_surface	303
building_height	190

Statistics. Table 1 summarizes the benchmark. The eight-task test set comprises 2932 questions over 421 locations across 40 cities; each location supplies one satellite tile and four street-level views.

4 Experimental Setup

Models. Our primary model is **Qwen3.5-9B** fine-tuned on the training split with LoRA [4] (rank 16, $\alpha = 16$, all linear layers including the vision tower), trained once per split and evaluated on the shared held-out benchmark. Fine-tuning on the task distribution is the cleanest setting for the view question, since a failure to use both views cannot then be blamed on poor task alignment. To check that the finding is not model-specific, we also evaluate four *off-the-shelf* remote-sensing VLMs zero-shot on the overhead half: GeoChat-7B [7], SkySenseGPT-7B [16], LHRs-Bot-Nova [10], and EarthDial-4B (RGB) [21], all single-image overhead-only models (Tab. 2).

Image-ablation protocol. Every question is evaluated under four input conditions: **blind** (prompt only), **sat** (the marked satellite tile only), **sv** (the four street-level views only), and **full** (satellite tile + four street-level views). The text of the question and options is **byte-identical** across conditions; only the set of supplied images changes. We decode greedily and parse a single answer letter. For the trained model we use the same image preprocessing as training (max edge 512 px).

Table 2: Off-the-shelf remote-sensing VLMs evaluated zero-shot on the overhead (satellite-only) half of the benchmark. All four are single-image and *overhead-only*; none ingests ground-level imagery. “OSM data” marks whether the model’s training corpus is derived from OpenStreetMap (relevant to the contamination caveat in Sec. 6).

Model	Base LLM	Vision encoder	Input res.	OSM data
GeoChat-7B [7]	Vicuna-1.5-7B	CLIP ViT-L/14	504	No
SkySenseGPT-7B [16]	Vicuna-1.5-7B	CLIP ViT-L/14	336	No
LHRS-Bot-Nova [10]	Llama-3-8B	SigLIP-L/14	336	Yes
EarthDial-4B-RGB [21]	Phi-3-Mini	InternViT-300M	tiled	Yes

Metrics. We report per-task accuracy in each condition and two derived quantities: $\text{fusion synergy} = \text{full} - \max(\text{sat}, \text{sv})$ and $\text{vision contribution} = \text{full} - \text{blind}$. Synergy > 0 indicates the second view adds accuracy a single view cannot; synergy near zero indicates redundancy. For the trained model we attach paired bootstrap 95% confidence intervals (10,000 resamples over questions) to the pooled synergy and the item-level quantities of Sec. 5.5.

Unified harness. All models, trained and zero-shot, run through one harness with a shared prompt, image rendering, and answer parsing, so the inputs are identical across models and only the forward pass differs; runs use a single RTX PRO 6000 GPU.

5 Results

5.1 The second view adds little over the better single view

Table 3 reports the trained Qwen3.5-9B on the held-out benchmark, per task, under all four conditions and both splits. First, full tracks the *better single view* closely: per-task fusion synergy ranges from -1.0 to $+4.5$ points and averages $+1.2/+1.3$ points (per-city / seen-unseen) across the eight tasks. The gain is positive but small, and street view alone (sv) already matches or exceeds the two-view full condition on land_use, road_type, and road_surface. Whether this margin reflects real fusion or the model defaulting to its stronger view is resolved at the item level (Sec. 5.5). Second, vision still matters: full – blind is large (for example $+22$ on transit_density and $+32$ on amenity_richness), so the model reads the imagery but extracts only a single view’s worth of information from it.

5.2 The same models fuse when fusion is required (positive control)

Could the small attribute-task synergy reflect an insensitive measurement rather than a real property of the tasks? Table 4 reports the same model and harness on the cross-view control tasks, whose answers require comparing the two views:

Table 3: Trained Qwen3.5-9B on the held-out benchmark, per attribute task and input condition (accuracy, %). **Syn.** = $\text{full} - \max(\text{sat}, \text{sv})$ is fusion synergy. Synergy is small on every task under both splits; full stays close to the better single view.

Task	per-city split					seen/unseen split				
	blind	sat	sv	full	Syn.	blind	sat	sv	full	Syn.
land_use	68.6	75.5	75.1	75.3	−0.2	63.4	72.0	72.4	73.9	+1.4
building_height	45.3	67.4	66.3	66.8	−0.5	50.5	63.7	66.3	66.8	+0.5
urban_density	54.6	61.8	62.7	65.1	+2.4	53.2	71.5	64.8	74.6	+3.1
junction_type	61.4	73.4	68.6	75.1	+1.8	62.0	74.0	71.3	76.3	+2.4
amenity_richness	26.4	53.4	56.5	58.7	+2.1	25.7	53.0	55.6	55.6	+0.0
road_type	68.6	72.4	77.4	76.5	−1.0	68.6	69.6	76.7	76.7	+0.0
road_surface	95.7	95.0	93.4	95.7	+0.7	95.7	95.0	95.4	95.0	−0.3
transit_density	28.3	43.7	45.6	50.1	+4.5	26.4	45.8	44.9	49.4	+3.6
Mean	56.1	67.8	68.2	70.4	+1.2	55.7	68.1	68.4	71.1	+1.3

Table 4: Positive control: the same trained Qwen3.5-9B (per-city) on cross-view tasks that require both views by construction. Synergy is large and positive, confirming the ablation detects fusion when it is needed.

Cross-view task	blind	sat	sv	full	Syn.
camera_direction	25.2	24.9	−	42.3	+17.3
mismatch_binary_easy	46.6	51.1	52.7	97.4	+44.7
mismatch_binary_hard	55.8	45.8	44.4	90.7	+44.9
mismatch_mcq_easy	20.7	26.4	23.5	96.2	+69.8
mismatch_mcq_hard	22.3	23.3	22.8	86.0	+62.7

synergy is large, +17 to +70 points, because one view is insufficient by construction. The same instrument reads a two-to-three-point gain on the attribute tasks and a forty-to-seventy-point gain when the task requires both views, so the small attribute synergy is a property of those tasks, not an insensitive test (Fig. 2).

That the cross-view tasks encode cross-view information, rather than being solved by some shortcut our VLM happens to exploit, is corroborated by an independent, purpose-built model. A cross-view retrieval network, Sample4Geo [1], transferred zero-shot and scored by cosine similarity (a different model class and metric, not comparable to Tab. 4), reaches 0.65 on the binary same-place/different-place questions—well above the 0.50 chance rate—confirming that the matching tasks carry genuine cross-view signal; the attribute tasks offer no comparable signal to recover.

5.3 The finding holds zero-shot across off-the-shelf RS-VLMs

To rule out that the redundancy is an artifact of our particular model, we evaluate four off-the-shelf remote-sensing VLMs zero-shot on the overhead (satellite-

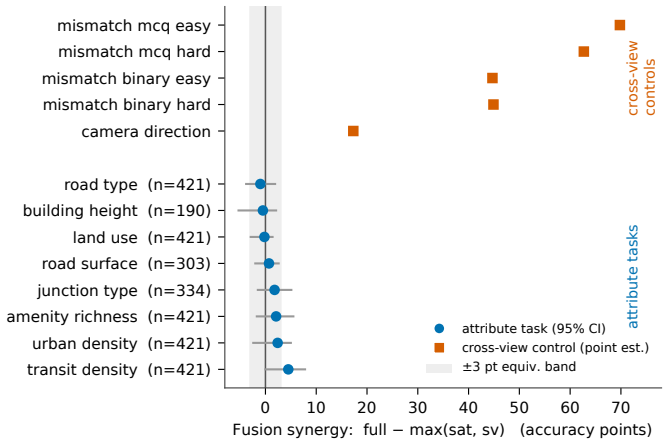


Fig. 2: Fusion synergy ($\text{full} - \max(\text{sat}, \text{sv})$) for the trained Qwen3.5-9B (per-city). The eight attribute tasks (blue, with paired bootstrap 95% CIs) cluster near zero—most CIs span the ± 3 -point equivalence band—while the cross-view control tasks (orange, point estimates) reach +17 to +70 points on the same axis and model. The instrument registers large synergy exactly when the task requires both views.

only) half of the benchmark (Tab. 5). None was trained on our task, our four-way MCQ format, or our marker overlay. They span CLIP- and SigLIP-based encoders, three LLM families, and single-crop versus tiled preprocessing. The trained Qwen3.5-9B in its satellite-only mode is the like-for-like single-overhead-view comparison; it outperforms the best zero-shot model (EarthDial, 39.9%) by +26.7 points overall, as expected for a model fine-tuned on the distribution. The four zero-shot models are satellite-only and therefore inform the single-view baseline and the contamination discussion (Sec. 6), not the synergy claim itself, which is a within-model comparison.

5.4 Stronger reasoning improves single-view reading, not fusion

If the two views *were* complementary but the model simply failed to combine them, a stronger reasoning procedure should surface the latent signal. It does not. Table 6 reports three strategies—plain decoding, a 16k-token explicit-thinking budget, and a four-turn *observe*→*reason*→*self-check*→*commit* conversation—on the eight attribute tasks, here using an AWQ-quantized Qwen3.5-9B so the long-context reasoning runs fit a single GPU. More deliberation raises accuracy modestly—the model gets better at each task in every view—but synergy stays *negative* throughout, and indeed *sv* alone is the single strongest condition under every strategy. The extra reasoning accrues to the single-view conditions just as much as to full, so what improves is the model’s reading of one view, not its ability to combine two. This mirrors the item-level finding of Sec. 5.5: the

Table 5: Off-the-shelf RS-VLMs, zero-shot, single marked satellite tile, per-task accuracy (%) over the eight-task benchmark ($n = 2932$). Last column: our trained Qwen3.5-9B in satellite-only mode (per-city), the like-for-like single-overhead-view reference. LHRS-Bot-Nova and EarthDial are trained on OSM-derived text and their scores should be read as an *upper bound* on visual skill (see Sec. 6).

Task	GeoChat	SkySenseGPT	LHRS-Nova	EarthDial	Ours (sat)
land_use	48.7	53.4	46.3	47.5	75.5
building_height	30.5	34.7	28.9	32.6	67.4
urban_density	13.5	14.5	17.8	30.4	61.8
road_type	10.7	18.1	26.1	46.1	72.4
road_surface	75.6	80.2	58.7	58.4	95.0
junction_type	8.4	30.8	35.3	52.7	73.4
amenity_richness	13.5	12.6	17.8	25.4	53.4
transit_density	21.4	26.1	34.2	29.7	43.7
Overall	26.2	32.0	32.4	39.9	66.6

Table 6: Prompting strategies on the eight attribute tasks (Qwen3.5-9B-AWQ, $n = 2932$, per-task means). **Syn.** = full-max(sat,sv); **Vis.** = full-blind. Reasoning improves accuracy but not synergy—the second view remains redundant, and street-view alone (sv) is the single strongest condition in every strategy.

Strategy	blind	sat	sv	full	Syn.	Vis.
plain	41.0	46.5	54.2	51.7	−2.8	+10.8
think-16k	41.4	48.2	54.0	52.5	−2.0	+11.2
multistep (4-turn)	43.4	50.3	55.0	53.2	−2.1	+9.8

obstacle is adjudicating between redundant views, which more thinking does not remove.

5.5 Item-level analysis: the second view interferes more than it helps

A small positive average could still hide genuine fusion on a subset of items. We join the predictions of all four conditions at the item level—every benchmark question carries a stable identifier shared across conditions—and analyze the 2932 fully paired eight-task items directly.

The second view overturns far more correct answers than it rescues. Comparing full against the better single view on each item is label-free and direct (Tab. 7). First, genuine fusion is rare: full is correct while *both* single views are wrong on only 26 items (0.9%, per-city) and 35 items (1.2%, seen/unseen). Second, interference is common: full is wrong while at least one single view was right on 298 items (10.2%) and 267 items (9.1%). The second view therefore

Table 7: Item-level analysis on the 2932 paired eight-task questions. **Fusion-win** = items full answers correctly while *both* single views fail; **Interfere** = items full fails while at least one single view succeeded. **synergy** = $\text{full} - \max(\text{sat}, \text{sv})$ and **full-oracle** (oracle keeps the better single-view answer per item; an upper bound on single-view routing, not an achievable system) carry paired bootstrap 95% CIs (10,000 resamples). The second view rescues few items and breaks many more.

Split	full	synergy	oracle	full-oracle	Fusion-win	Interfere
per-city	69.5	+2.2 [0.9, 3.3]	78.8	-9.3 [-10.4, -8.2]	0.9%	10.2%
seen/unseen	70.3	+2.8 [1.5, 3.7]	78.2	-7.9 [-9.1, -6.8]	1.2%	9.1%

overturns a correct answer roughly ten times as often as it rescues a wrong one. A McNemar test on the paired full-versus-single outcomes confirms the imbalance is not noise ($p < 10^{-3}$ for full vs. either single view on both splits; e.g. 219 vs. 133 discordant items for full vs. **sat**, per-city, $p = 5 \times 10^{-6}$). When the two views disagree, the model does not reliably keep the correct one; the extra view is, on net, a distractor.

An oracle single-view router would beat the two-view model. The same asymmetry can be stated as an upper bound. Consider an *oracle* that, per item, keeps the better of the two single-view answers—a router, not a fusion model, and not an achievable system, since selecting the right view uses the gold label. This upper bound on single-view *routing* exceeds full by 9.3 points (per-city) and 7.9 points (seen/unseen), with paired bootstrap 95% intervals that exclude zero ($[-10.4, -8.2]$ and $[-9.1, -6.8]$, 10,000 resamples; Tab. 7 and Fig. 3).

Aggregate synergy is small and positive—not zero. The average gain is not zero. Table 7 reports the pooled fusion synergy with its bootstrap CI: +2.2 [0.9, 3.3] (per-city) and +2.8 [1.5, 3.7] (seen/unseen); both intervals sit above zero. (These item-weighted pooled values slightly exceed the +1.2/+1.3 unweighted task means of Tab. 3 because the larger tasks carry the larger synergies.) An equivalence test (TOST, [8]) against a ± 3 -point margin does *not* declare the synergy negligible—it is a real, small, positive effect. Our claim is correspondingly bounded: the second view adds a couple of points on average, more than an order of magnitude below the +17 to +70 points it adds on the cross-view controls (Tab. 4), while breaking roughly one in ten items it touches. Whether a ~ 2 -point average gain at that cost warrants serving a second imagery modality is the practical question we take up in Sec. 7.

6 Robustness and Confounds

Option text and position. Because the questions are templated, we check that the answer cannot be read off the options. The correct letter is close to

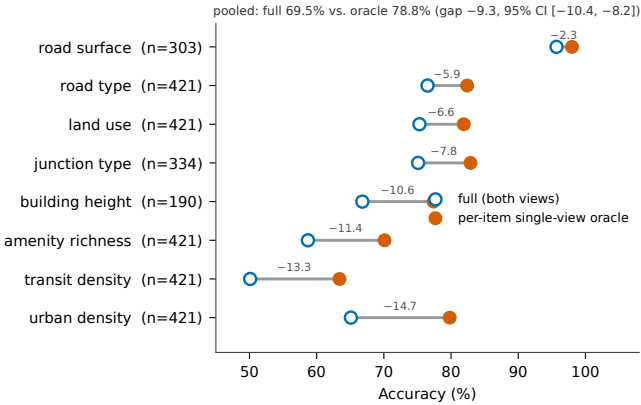


Fig. 3: Per task, the two-view full condition (open circles) versus a per-item single-view oracle that keeps the better single-view answer for each question (filled circles; an upper bound on single-view routing, not an achievable system). The oracle leads on every task. Trained Qwen3.5-9B, per-city.

uniform across the eight retained tasks (A/B/C/D = 743/766/703/720; $\chi^2 = 3.08$, $p = 0.38$ against uniform), and the correct option is not systematically the longest—its mean length (24.0 characters) is slightly below that of the distractors (25.0), and it is the single longest option on only 19.8% of items, below the 25% chance rate. The blind accuracy we report per task (Sec. 5.1) is the empirical ceiling on everything guessable without images and subsumes these effects.

Text-only leakage and degenerate tasks. We exclude the one task that was fully text-solvable (Sec. 3). One retained task, road surface, is near ceiling and almost text-solvable (blind = full = 95.7, vision contribution ≈ 0), so its small synergy reflects that nothing in the image matters, not redundancy between views. Restricting the headline finding to the seven tasks with a meaningful vision contribution (full – blind > 5 points) leaves the conclusion unchanged: mean synergy is +1.5 points ($n = 2629$). We therefore rank and interpret tasks by vision contribution rather than raw accuracy throughout.

OSM provenance. Our labels are OSM-derived. Two of the four zero-shot models—LHRS-Bot-Nova and EarthDial—are trained on OSM-derived text (LHRS-Align captions OSM tags across 9259 cities; EarthDial’s pretraining QA is built from OSM-labeled imagery), so their scores may reflect shared OSM priors, and we treat them as an upper bound on visual skill; GeoChat and SkySenseGPT do not share this provenance. The synergy result, however, is a *within-model difference* (full vs. the better single view), so any view-independent OSM prior cancels in the difference and cannot produce the effect.

Satellite resolution. The redundancy is measured against high-resolution overhead imagery, not a coarse proxy: every benchmark tile resolves to a sub-meter source (ESRI ~ 0.3 m/px to 0.5 m/px, NAIP ~ 0.6 m/px, IGN ~ 0.2 m/px); the Sentinel-2 fallback (~ 10 m/px) is never triggered. For tasks that are structurally ground-level (road surface, signage-driven amenities) no overhead resolution recovers the cue. We cannot rule out that still higher ground sampling distance would help the few tasks where overhead geometry is plausibly resolution-limited (building height, urban density); we flag this as the main open question in Sec. 7.1.

Image count and attention. The street-view condition supplies four images and the satellite condition one, so part of street view’s edge could be image count rather than viewpoint. Three points bound this concern. First, our headline result does not require `sv > sat`: it is stated against the *better* single view, whichever that is, and the interference asymmetry holds regardless. Second, an attention check argues against a pure token-count effect—on a separate attention-instrumented model, the raw cross-modal attention at the answer token favors street view (6.3% vs. 2.0%), but *per image* the satellite tile draws *more* attention than any single street view (24.3% vs. 18.9%), so the satellite tile is read at least as much as one street view (attention is reported only as corroboration; its limits as explanation are well known [6, 24]). Third, our recommendation is to serve the cheaper single view, so street-view richness is part of why one view suffices. We did not, however, equalize image count directly by comparing a single street-level angle to the four-view grid, and flag that as a limitation (Sec. 7.1).

7 Discussion

Our reading is that complementarity is a property of the task rather than of imagery in general. An attribute question, such as the local land use, building height, or road surface, can be answered from one location’s imagery, and the two views then encode much the same answer-relevant content, so the model reads it off whichever view is more convenient. A cross-view question (whether a street photo belongs to a given tile, or which way a camera points) instead depends on the relationship between the views, where our models do use the second view. The eight attribute tasks fall in the first regime, the cross-view tasks in the second.

This has a practical consequence. Street-level and high-resolution satellite imagery differ sharply in cost, coverage, and licensing, yet for these attribute families the second view adds only two to three points and naively degrades about one item in ten. A deployment can serve a single view, the cheaper or better-covered one, at little accuracy cost; and when both are available, the evidence favors *routing* each question to one view over feeding the model both. The synergy, vision-contribution, and interference measures we use form a reusable instrument for making that call before committing to dual-view collection.

7.1 Limitations and scope

The claim concerns eight urban-attribute tasks in four-way MCQ form at commonly served satellite resolutions; it is not a claim that overhead imagery is uninformative or that fusion fails in general, as the cross-view controls show large synergy on the same models. Four caveats bound it further: we did not equalize image count between the single-view conditions (Sec. 6); our two splits are two training regimes on one shared held-out test set, of which the held-out-cities split is the true generalization check; higher ground sampling distance might help the few overhead-geometry-limited tasks (building height, urban density); and an open-ended or regression formulation could surface complementary signal an MCQ cannot express.

8 Conclusion

We measured, rather than assumed, the value of a second perspective for urban attribute recognition. Across model scale, training regime, city split, and prompting strategy the second view adds only a few points over the better single view and overturns a correct answer about ten times as often as it rescues a wrong one, while the same models gain 17–70 points on cross-view tasks that require both views. Multi-perspective necessity is thus task-dependent; for these urban-attribute tasks, a single well-chosen view might be enough.

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